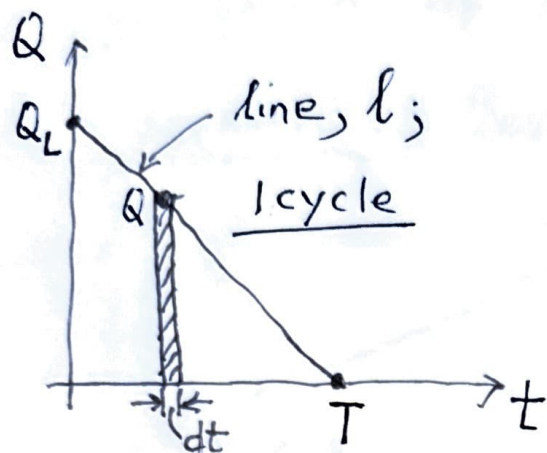


Agenda

1. Cycle or Average Inv. & Flow time
2. Aggregation of cycle inventory
 - 2.1 No aggregation
 - 2.2 Bridge
 - 2.3 Simple aggregation

1. → Average Inventory

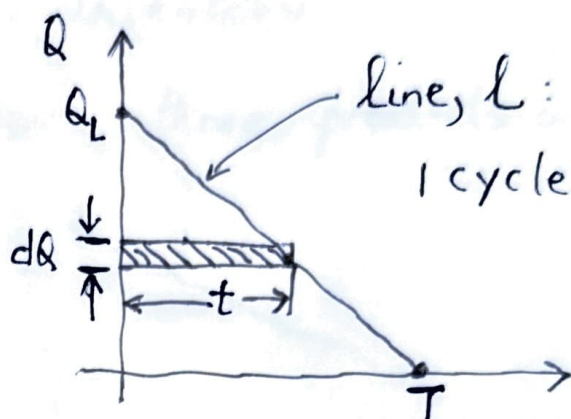


$$\begin{aligned} \text{Average inventory} &= \frac{1}{T} \int_0^T Q dt \\ &= \frac{1}{T} (\text{area under line, } l) \\ &= \frac{1}{T} \left[\frac{1}{2} Q_L T \right] \\ &= \frac{Q_L}{2} \end{aligned}$$

$$\text{Cycle inventory} \triangleq \text{average inventory} = \frac{Q_L}{2}$$

Flow-time

11.2

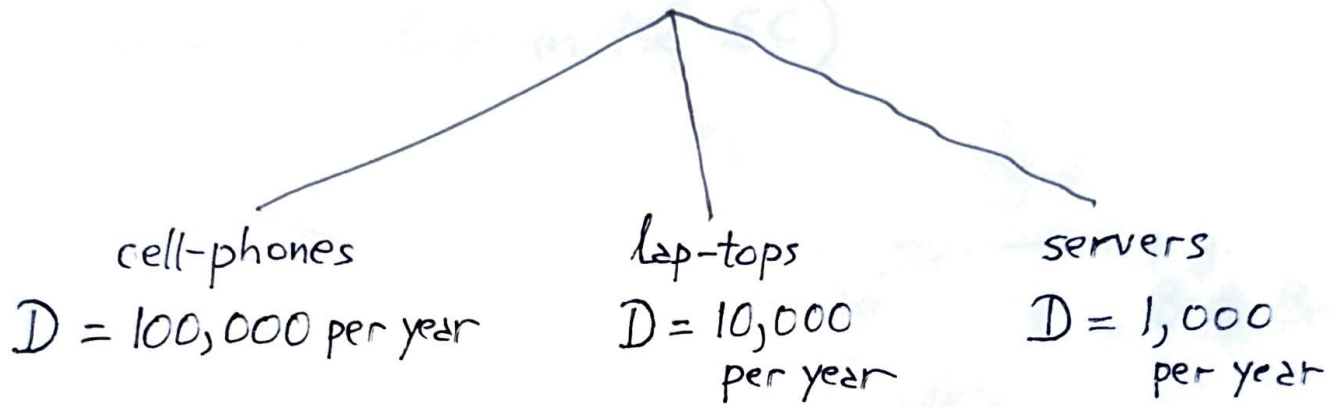


$$\begin{aligned} \text{Average time} &= \frac{1}{Q_L} \left[\int_0^{Q_L} t dQ \right] \\ &= \frac{1}{Q_L} (\text{area under line, } l) \\ &= \frac{1}{Q_L} \left[\frac{1}{2} Q_L T \right] \end{aligned}$$

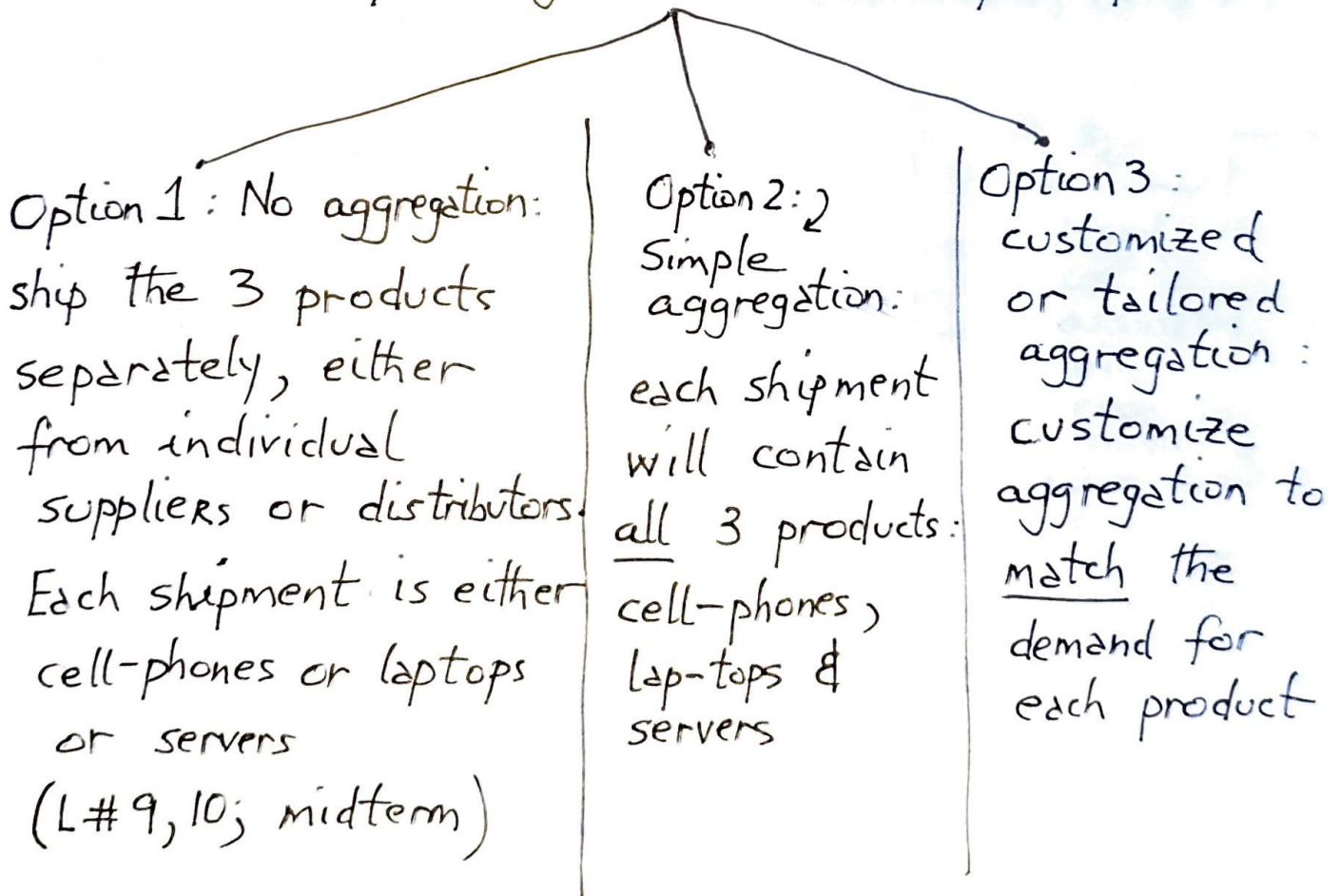
$$\text{flow time} \triangleq \text{average time} = \frac{T}{2}$$

2. Aggregation of cycle inventory

Scenario: Best Buy: three products of interest

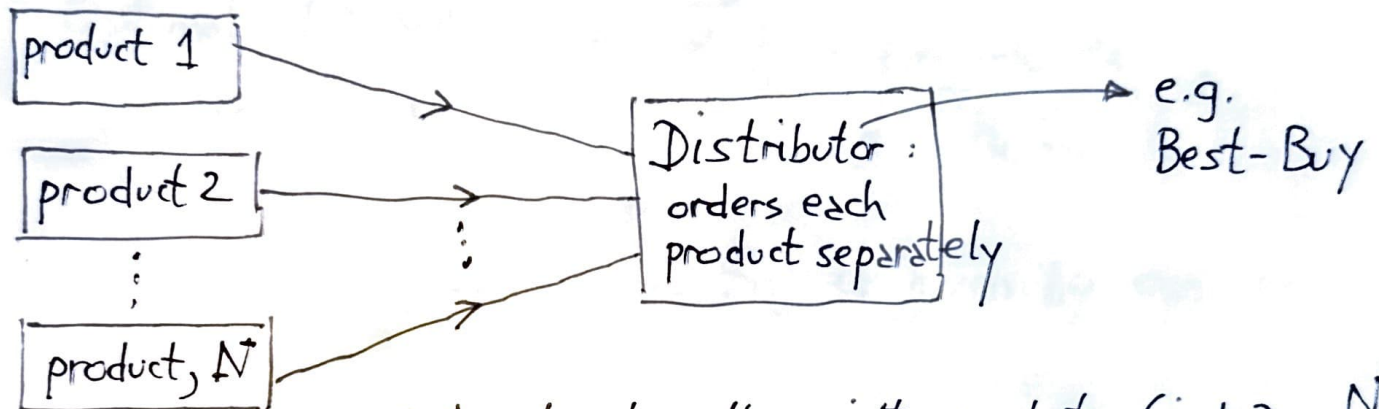


Cycle Inventory Management for Best-Buy: options



2.1 Case 1 : Multiple products , no aggregation

Each product is shipped separately (from the preceding stage in the SC)



i^A denotes the i th product, $(i=1,2,\dots,N)$

Fixed order cost for each product, i , $S_i = \underbrace{S}_{\substack{\text{common} \\ \text{base} \\ \text{ordering} \\ \text{cost}}} + \underbrace{s_i}_{\substack{\text{fixed cost} \\ \text{associated} \\ \text{with} \\ \text{each} \\ \text{product} \\ \text{aka} \\ \text{product} \\ \text{specific} \\ \text{cost}}} \rightarrow (1)$

$$(i=1,2,\dots,N)$$

Since each product is being shipped separately,
 the results are the same as for a single product
 [studied & derived in L# 9, 10; and used on the midterm]

$$\Rightarrow \left. \begin{array}{l} \text{Optimal lot size} \\ \text{for each product, } i \end{array} \right\}, (Q_L^*)_i = \sqrt{\frac{2 D_i S_i}{h_i C_i}}, \rightarrow (2)$$

$(i=1, 2, \dots, N)$; & S_i is given by eqn (1)

$$\left. \begin{array}{l} \text{Optimal order frequency, } n_i^* \\ \text{for product } i \end{array} \right\} = \frac{D_i}{(Q_L^*)_i}, \rightarrow (3)$$

$(i=1, 2, \dots, N)$

... ETC.

(See L# 10 for the catalog of eqns.)

2.2. Bridge from option 1 to option 2

In L # 9, 10 we saw that for a single product,

$$\text{total cost, } C_T = f(Q_L), \longrightarrow (1)$$

& then set $\frac{dC_T}{dQ_L} = 0$, to obtain the optimal lot size, Q_L^*
(see L # 10)

We could also formulate the optimization problem in terms of the shipment or order frequency, "n", using

$$n = \frac{D}{Q_L} \longrightarrow (2)$$

We can write,

$$C_T \underset{(1,2)}{=} f\left(\frac{D}{n}\right) \triangleq g(n),$$

$$\& \text{ Then set } \frac{dC_T}{dn} = \frac{dg(n)}{dn} = 0, \longrightarrow (3)$$

to obtain the optimal shipment or order frequency, n^*

$$C_T = \underbrace{C_M}_{\text{material cost}} + \underbrace{C_S}_{\text{order or shipment cost}} + \underbrace{C_I}_{\text{cycle inventory holding cost}} \rightarrow (4)$$

total annual cost

$$= DC + nS + \left(\frac{Q_L}{2}\right)(hC) \rightarrow (5)$$

$$n Q_L \stackrel{(2)}{=} D, \text{ or } Q_L = \frac{D}{n} \rightarrow (6)$$

$$C_T \stackrel{(5,6)}{=} DC + nS + \left(\frac{D}{2n}\right)hC \rightarrow (7)$$

$$C_T = \underbrace{\hspace{10em}}_{g(n)}$$

To find the optimal shipment frequency, n^* , that minimizes the total cost, C_T

$$\frac{dC_T}{dn} \stackrel{(7)}{=} 0 + (1)(S) + \left(\frac{-D}{2n^2}\right)hC = 0$$

to find the minimum

$$\Rightarrow n^* = \sqrt{\frac{DhC}{2S}} \rightarrow (8)$$

checking eqn. (8)

$$Q_L^* = \frac{D}{h^*} \sqrt{\frac{2DS}{hC}},$$

which is the eqn. for Q_L^* that was derived in L#10

2.3. Cycle inventory aggregation, Option 2:

Simple aggregation:

Multiple products, simple aggregation

⇒ Each shipment (or order) contains all products. The lot size for each product will be different.

Example:

No aggregation:

1 truckload contains

100 refrigerators (say)

or

1000 laptops (say)

or

10,000 cell-phones (say)

Simple aggregation:

1 truckload contains

50 refrigerators (say)

and

500 laptops (say)

and

5000 cell-phones (say)

Option 2 : Simple aggregation

If every order (or shipment) contains all products, then the order (or shipment) frequency, "n", is the same for all products.

Each product, however, will have a different lot size, $(Q_L)_i$, $(i = 1, 2, \dots, N)$

Problem : Determine the optimal shipment (or order) frequency, n^* , that minimizes the total cost, C_T .

Process

$$\text{Order (or shipment) cost} = S^* = \underset{\substack{\uparrow \\ \text{common} \\ \text{base} \\ \text{ordering} \\ \text{cost}}}{S} + \sum_{i=1}^N \underset{\substack{\uparrow \\ \text{product} \\ \text{specific} \\ \text{cost for} \\ \text{product } i}}{s_i}, \rightarrow (1)$$

$i = (1, 2, \dots, N)$; $N \triangleq$ total # of products.

$$C_M = \text{annual material cost} = \sum_{i=1}^N D_i C_i \rightarrow (2)$$

↑
unit cost

$$C_S = \text{annual shipment cost} = n S^* \rightarrow (3)$$

$$C_I = \text{cycle inv. holding cost} = \sum_{i=1}^N \left(\frac{Q_i}{2} \right) h_i C_i$$

$$= \sum_{i=1}^N \frac{D_i h_i C_i}{2n} \rightarrow (4)$$

$$\text{Total annual cost, } C_T = C_M + C_S + C_I$$

$$C_T \stackrel{(2,3,4)}{=} \sum_{i=1}^N D_i C_i + n S^* + \sum_{i=1}^N \frac{D_i h_i C_i}{2n} \rightarrow (5)$$

To find the optimal order frequency, n^*

$$\frac{dC_T}{dn} \stackrel{(5)}{=} 0 + (1) S^* + \sum_{i=1}^N \left(\frac{D_i h_i C_i}{-2n^2} \right) = 0$$

to find
the minimum

$$\Rightarrow n^* = \sqrt{\sum_{i=1}^N \left(\frac{D_i h_i C_i}{2 S^*} \right)} \rightarrow (6)$$